

Hate Crimes in the United States: A study of causes and trends

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Project Introduction

Background

Hate crimes have a long and painful history in the United States, often reflecting broader social tensions and structural inequalities. From racially motivated violence during Reconstruction, to attacks against immigrants in the early 20th century, to the targeting of LGBTQ+ communities, hate crimes have evolved alongside America's shifting political and cultural landscape. Over time, federal and state governments have made efforts to address these acts through legislation, notably beginning with the Civil Rights Act of 1968 and expanding through laws like the Hate Crimes Prevention Act of 2009. Despite these measures, hate crimes remain a persistent and complex societal issue.

Recent data reveal troubling trends: not only have hate crime rates risen in the past decade, but the nature and motivation behind these crimes have also diversified. Factors such as race, religion, sexual orientation, gender identity, and ethnicity continue to drive targeted violence. Our analysis of national hate crime data highlights the uneven implementation of hate crime legislation across states, raising important questions about the relationship between legal protections, reporting practices, and actual rates of victimization.

Throughout history, attempts to mitigate hate crimes have ranged from expanding legal definitions of bias-motivated crimes to strengthening reporting mechanisms and improving victim services. Yet, challenges remain in ensuring equitable protection across communities and states. Why do hate crimes occur, and why are some groups disproportionately targeted? How do political, cultural, and demographic factors influence both the prevalence of hate crimes and the efficacy of policies meant to prevent them? Understanding these questions is critical, not just for academic inquiry, but for designing effective, compassionate interventions that safeguard vulnerable communities.

Our project aims to investigate these dynamics through a data science lens. Specifically, we explore how the presence and implementation of hate crime legislation affects hate crime rates and victim protections across politically diverse states, and what factors contribute to the efficacy of such policies regardless of a given state's political climate.

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By modeling the relationships between legislation, demographic trends, and hate crime occurrences, we hope to identify actionable insights that can inform both policy and public awareness. In doing so, we seek to deepen our collective understanding of hate crimes—not only why they happen, but how we can meaningfully work to prevent them.

Research Question

How does the presence and implementation of hate crime legislation affect hate crime rates and victim protections across politically diverse states, and what factors contribute to policy efficacy regardless of a given state's political climate?

Why this question is important?

Given the large-scale violence and impact that hate crimes can have, we wanted to examine how effective (a) the presence of legislation aiming to prevent these incidents has been and (b) the extent to which legislation exists (stricter vs. more lenient) has proven helpful. We thought that if we could uncover significant findings about the efficacy of legislation, we could know whether it would be helpful for other states to stricken theirs or whether a different approach needs to be taken to prevent hate crimes altogether due to a lack of success of current legislation.

How we answered this question?

We answered this question by researching each state's level of legislation and creating our own relative scale to measure how "extensive" depending on how many bias categories are protected. First, we explored the data by visualizing correlations of the extent of legislation to the number of recorded hate crimes in a state, the number of hate crimes per state over time, the number of hate crimes over time per level of legislation, and the most frequent biases exhibited by state. Then, we gathered our findings and created models to address various parts of our question. We created a multinomial logistic regression to predict which bias categories would be safe within a state given its level of legislation – this showed us how effective legislation actually was at protecting each bias category within a given state. Next, we created a multi-class logistic regression to predict which state a given hate crime occurred in – this analysis showed us what indicators (certain groups, races, etc.) were most frequently correlated with hate crimes in specific states and should be targeted with future legislation. Finally, we created a random forest, logistic regression, and XGBoost model to predict the motivation behind a given hate crime – similar to earlier, this could provide us with information on what motivations need to be better protected in states.

EDA & Takeaways

Data applicability and ethical data collection standards

Data Applicability to Research Question:

- The hate_crimes dataset provides a comprehensive overview of hate crime incidents across the United States, capturing detailed information on the location, type of offense, and characteristics of offenders and victims. We also modified the dataset and added features like level of hate crime legislations across states in our dataset. This dataset is highly relevant to the research question, as it allows for an analysis of how different levels of hate crime legislation (none, some, extensive) correlate with hate crime rates and victim protections. Additionally, the dataset includes state-level information, which can be used to examine the impact of political affiliation on hate crime legislation and its efficacy.

Ethical Data Collection Standards:

- The dataset appears to adhere to ethical data collection standards by providing anonymized and aggregated data on hate crime incidents, ensuring that individual victims and offenders cannot be identified. This protects the privacy of those involved and minimizes potential harm. Also, the data sources used to compile this dataset are reliable and the data collection methods are transparent. Finally, the dataset captures a wide range of hate crime types and demographics, providing a comprehensive view of hate crime incidents across the United States. This inclusivity helps ensure that the analysis is representative and does not disproportionately exclude certain groups or regions.

Correlation Matrix

We decided to construct a few correlation matrices in order to understand how some variables relate to each other for the purpose of our question. We wanted to examine if legislation level was correlated in any way with hate crime counts or if specific crime types were correlated with increased or decreased hate crime counts.



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Legislation Level

Correlation between LEGISLATION_LEVEL and TOTAL_CRIMES: The correlation coefficient is 0.20. This indicates a weak positive correlation between the legislation level and the total number of crimes. There can be a mutual causation here: a state needs to have some legislations so they would record the hate crimes incidents; at the

same time, as there are more hate crimes in a state, the state may pass more legislations protecting different groups.

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The positive correlations with extensive legislation suggest that higher numbers of incidents might encourage more extensive legislation. This indicates that legislation is responsive to the needs highlighted by the incidence of crimes. Also, it may indicate that more legislations makes the recording process of hate crimes more effective and regulated so we can see more records. Therefore, we think there are some looping effects that are important here.

The correlations for "Some" legislation are mixed, with both positive and negative values. The negative correlations might suggest that in areas with some legislation, there is a slight tendency for lower crime counts for certain types of crimes, but again, these are not strong correlations.

The correlations for "None" legislation are all negative, albeit weak.

Number of Hate Crimes per State Over Time Choropleth Map

By grouping the dataset by state and year, we were able to get an aggregate number of hate crimes that occurred annually by state. Using Folium, we were able to create a choropleth map that visualizes the changes in frequency over time.

From this visualization, we observed that surprisingly large states like California/Texas did not always having the highest number of hate crimes recorded, which we didn't expect given their large size within which they could occur, diverse populations that could have spite against one another, and overall high populations which could have a higher probability of offenders. As for year-over-year trends, states in the South (Mississippi, South Carolina, Georgia, etc.) consistently experience high relative annual levels of crime alongside some of the central states (North/South Dakota, Montana, Utah). From what we know about the demographic breakdowns of these states, hate crimes occurring more frequently in these areas makes sense – they are predominantly right-leaning areas without much racial diversity.

Type of Biases Facet Wrap

We clarified these three categories with no legislation having 0 bias categories protected (essentially no hate crime legislation exists), some legislation having 1-4 bias categories protected, and extensive legislation having 5-7 bias categories protected. These bias

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categories include Race/ Color, National Origin, Religion, Sexual Orientation, Gender/ Sex, Gender Identity, and Disability.

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The graphs above illustrate trends in hate crime counts grouped by states that have no, some, and extensive hate crime legislation. It is important to note, however, that hate crime legislation was broadly enacted in 2019, which is not a date captured in the data used to construct the above illustration. Given this, the graphs above depict that states privy to a significant count of hate crimes tend to adopt hate crime legislation. Higher crime counts may drive more legislation. At the same time, extensive legislation may improve crime recording, while some or no legislation may not significantly impact crime counts. Overall, our findings highlight potential looping effects between crime incidence and legislative response.

Mapping Hate Crime Bias Counts by State Choropleth Map

Next, we grouped the dataset by the aforementioned bias categories to find that 'Race/Color' was the most frequent type of hate crime that occurred across nearly every state except New York as the first Folium visualization shows. We also noticed from this that more hate crimes incidents across most bias types typically has a positive correlation with the legislation level.

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To further analyze incidences of 'Race/Color' hate crimes, we further grouped the data by type of 'Race' which is displayed in the second Folium visualization. We found that the most common type of racial bias hate crimes is highly correlated with state demographics. In most states, "Anti-Black or African American" is the most frequent type. However, in states with distinct demographic compositions, the most common type may vary.

EDA Takeaways

In our exploratory data analysis (EDA) of the hate crimes dataset, we uncovered a lot of insights that helped shape how we thought about modeling the relationship between hate crime legislation and hate crime rates. The dataset had detailed, incident-level data across U.S. states—including offense types, offender and victim demographics, and a new variable we created that categorized each state by its level of hate crime legislation (none, some, or extensive) using the level of bias categories protected.

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After cleaning and filtering the data to focus on state-level trends, we noticed that hate crime counts were both overdispersed and zero-inflated—basically, many states reported zero incidents in certain years, especially those with little or no hate crime legislation in place. There also seemed to be regional patterns—Southern and some Central states consistently had higher hate crime counts, which suggested to us that regional or spatial effects had to be considered.

One thing that stood out was how much variation there was in the types of biases motivating these crimes. Across almost every state, race or color-based hate crimes were the most common, but New York was the only exception—religious bias was most prominent there.

When we looked at correlations, we found that states with more extensive hate crime legislation tended to report higher counts across most bias categories. We thought that this could potentially be suggesting a feedback loop where higher levels of hate crime lead to more legislation having to be put into place, which in turn leads to better reporting and detection of incidences. Using all of this information and preliminary analysis, we were able to move forth into modeling our dataset to uncover further revelations.

Modeling & Modeling Analysis

Data Cleaning Procedure

To clean the data, we started by dropping features that were majority null ("PUB_AGENCY_UNIT", "ADULT_VICTIM_COUNT", "JUVENILE_VICTIM_COUNT", "ADULT_OFFENDER_COUNT", "JUVENILE_OFFENDER_COUNT", "OFFENDER_ETHNICITY") and any null values in the dataframe as well since this only dropped a few rows. Then, we narrowed down our features to include 'AGENCY_TYPE_NAME', 'STATE_ABBR', 'POPULATION_GROUP_DESC', 'INCIDENT_DATE', 'TOTAL_OFFENDER_COUNT', 'OFFENDER_RACE', 'VICTIM_COUNT', 'OFFENSE_NAME', 'TOTAL_INDIVIDUAL_VICTIMS', 'LOCATION_NAME', 'BIAS_DESC', and 'VICTIM_TYPES'. The 'STATE_ABBR' feature was kept as it was our target classification for the model and all other features remaining can help train the model for its prediction. Next, we created dummy variables to handle categorical data for the 'AGENCY_TYPE_NAME', 'POPULATION_GROUP_DESC', 'OFFENDER_RACE', 'MULTIPLE_OFFENSE', 'MULTIPLE_BIAS', and 'INCIDENT_MONTH' features. There were a few features that needed to be handled a little differently. For example, with 'INCIDENT_DATE' we sought out just the month value to be able to one-hot-encode this feature, resulting in the 'INCIDENT_MONTH' feature. For the sake of 'BIAS_DESC', 'PUB_AGENCY_UNIT', 'ADULT_VICTIM_COUNT', 'JUVENILE_VICTIM_COUNT', 'ADULT_OFFENDER_COUNT', 'JUVENILE_OFFENDER_COUNT', 'OFFENDER_ETHNICITY', 'AGENCY_TYPE_NAME', 'POPULATION_GROUP_DESC', 'OFFENDER_RACE', 'MULTIPLE_OFFENSE', 'MULTIPLE_BIAS', and 'INCIDENT_MONTH' features, there were some values

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that contained multiple categories, so we created a dummy variable for every combination. After this lengthy cleaning procedure, our data was ready to be modeled.

Predicting Which Bias Categories are Protected by a State's Legislation

Multinomial Logistic Regression

We chose a multinomial logistic regression model to predict which bias categories are protected by state legislation because its typically good at predicting nominal (categorical) dependent variables with more than two categories. The independent variables that we will be using are the count of hate crimes that occurred by state, a bias description of each hate crime, and offender demographics. We think that it would be applicable to this dataset because it could predict bias categories (e.g., Race, Religion, Sexual Orientation) using ordinal predictive features such as 'Legislation_Level' which only contains values of 'none', 'some', and 'extensive'. From our previous EDA, we found that legislation unevenly correlates with bias type, so this model will be assessing how legislative strength shifts the likelihood of protecting specific groups through odds ratios to answer the section of our research question focusing on policy efficacy.

When we evaluated the model's performance, the overall accuracy was about 30%, which might seem low at first glance. But since this was a multi-label classification problem—where each state could have multiple protections—that level of accuracy isn't unusual. Tying this back to our EDA, it's clear the model is picking up on the more common patterns—Race, Religion, and National Origin are widely protected and well-represented in the training data. But for categories like Gender Identity and Disability, which are less frequently protected, the model struggled. Even though we found a positive correlation between stronger legislation and the number of protected classes, the model wasn't able to capture those rare cases very well.

Predicting Which State A Given Hate Crime Occurred In

Multi-Class Logistic Regression

We chose a multi-class logistic regression model to predict which state a given hate crime took place in because it could provide us with additional information on which features in the dataset point to certain states (e.g., groups of people that are more likely to commit or be targeted by hate crimes). The independent variables that we will be using are the agency type, population level, incident date, total offender count, offender victim count, name of offense, location, type of bias, and type of victims (e.g.,

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group, individual). Not only does this model tell us the most common type of feature per state, but also how much that feature indicates a specific state, if at all.

Class	Precision	Recall	F1-score	Support
Race/Color	0.90	1.00	0.95	9
National Origin	0.90	1.00	0.95	9
Religion	0.90	1.00	0.95	9
Sexual Orientation	0.75	0.75	0.75	8
Gender/Sex	0.80	1.00	0.89	8
Gender Identity	0.00	0.00	0.00	2
Disability	0.67	0.86	0.75	7

The model achieved a 30% accuracy for classification, but given that the dataset is biased towards states with larger numbers of reported hate crimes and that there are 50 possible classifications to be made, this is not an unusually low metric. The states that had a greater number of incidents reported also had higher accuracies, so we assumed that if there was more data for some states with lower accuracies, this metric could increase. Regardless, we were able to understand which features were most prevalent across states (offender demographics, victim demographics, location, etc.), which can provide insight for what future legislation should target to successfully deter hate crimes, or where existing legislation is lacking.

Predicting the Motivation Behind A Hate Crime Case

Logistic Regression

We chose a logistic regression model to predict the motivation behind a certain hate crime instance because it can provide us with preliminary insights on this classification task due to its nature of being simple with a low risk on overfitting with a relatively simpler dataset, as we consider ours to be. The independent variables that we will use are the year of the instance, the state in which it took place, the level of legislation in the state, type of offense, population level, total offender count, offender race, victim count, type of victims (e.g., group, individual), and whether this was the offender's first offense. Since there were some rows in the dataset that had multiple motivations linked to them, we narrowed down our predicted values to just predict one from the following: 'Race/Color', 'Religion', 'Sexual Orientation', 'Disability', 'Gender Identity', 'Gender/Sex'.

Our results suggest that different types of bias-related hate crimes exhibit distinct geographic, demographic, and situational patterns. For example, with the motivations of 'Race/Color' and 'Religion', the presence of state legislation and the state itself (e.g., Ohio,

New York) served as strong predictors; we think this means that reporting practices vary by jurisdictional policy environments which had an impact on our model's classifications. Relating back to our research question, we found that the presence of hate crime legislation is associated with a higher likelihood of certain bias types (e.g., disability or religion) being reported or classified as such, particularly in states with partial or extensive legislation. We think that legislation can help perpetrate reporting and clear classification of hate crimes, regardless of a state and its population. Additionally, differences across states and urban/rural settings imply that political and demographic contexts influence how legislation operates, pointing towards the importance of implementation and institutional engagement beyond just passing laws.

Random Forest

We also chose a random forest model for this predictive task because these are well-suited to handle nonlinear interactions with complex features – both categorical and continuous – that our logistic regression model could not have captured. We used the same independent variables as the logistic regression model, as well as the same predictive requirements for only having one motivation per case.

Class	Precision	Recall	F1-score	Support
Disability	0.33	0.13	0.18	311
GenderIdentity	0.21	0.07	0.11	111
Gender(Sex)	0.00	0.00	0.00	47
Race(Color)	0.71	0.84	0.77	25,700
Religion	0.55	0.41	0.47	7,280
SexualOrientation	0.32	0.18	0.23	6,208

The model achieved a 65% accuracy for classification. Most notably, it's able to perform extremely well in cases of 'Race/Color' bias which also happen to be the most frequently observed motivation in the dataset (25,000 cases out of 40,000). However, for minority classes such as 'Gender Identity' or 'Disability' that had less instances in the dataset, the model either misclassifies or doesn't predict instances to fall into these categories at all. In a real-life application, we think this model could be very helpful in analyzing motivations if the minority classes are oversampled with weights to account for the dataset's imbalance.

XGBoost

Finally, we chose an XGBoost model for this question to further attempt at predicting motivations, but try to account for the large class imbalance that was presenting weaker results in previous models. XGBoost learns interactions naturally from the data without

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assuming linearity, similar to Random Forest, but also prevents overfitting on more frequently observed classes which is beneficial for a dominated dataset like ours. The independent variables and predictive nature of the model are the same as for the other two models for this task.

The model's accuracy was slightly higher than the Random Forest model, but did not address the issue of class imbalance as well as hoped. 'Race/Color' accuracy outperformed that of previous models, however, less frequently observed categories still had very accuracies as a result of misclassifications or lack of classifications altogether.

Modeling Takeaways

Overall, our models tended to perform well on common bias categories, especially 'Race/Color', but struggled with underrepresented ones like 'Gender Identity'. This was a recurring trend across our models, so moving forward we will definitely have to account for class imbalance for more accurate models. Outside of this, predictions were frequently influenced by certain states, urban levels, and victim types. These may not be some of the strongest predictors (ex. states), which raises concerns about the feature importances that the models picked up on. Moving forward, we hope to build and improve the aforementioned shortcomings of our models.

Conclusion

Our research question focused on addressing how the presence and enforcement of hate crime legislation in a state affect hate crime rates and victim protections, and what factors contribute to the effectiveness of these policies? As detailed earlier, we began by creating visualizations to better understand the data – how correlated legislation levels were with the number of hate crimes that took place in a state, the number of hate crimes per state over time, the most frequent biases of hate crimes per state, and trends in hate crime frequency depending on legislation level. To address our question, we found that states experiencing higher numbers of hate crimes typically also have higher levels of legislation.

We further analyzed the parts of our question with various models. With multinomial logistic regression, we attempted to predict the number of bias categories protected by state where we found a positive correlation between strong legislation and the number of protected classes, however, the model struggled to predict less frequently protected categories. Next, we performed multi-class logistic regression to predict what state a given hate crime took place in. This analysis gave us the strongest predictive features per state; this knowledge could be useful to gauge whether existing policies address each of these features and, if not, enact future legislation to do so. Finally, we used multinomial logistic regression, random forest, and XGBoost to predict the motivation behind a

specific hate crime. We found that this model most frequently predicted a motivation of 'Race/Class' because of the prevalence of this classification in the data, while it struggled to predict any of the minority classes due to their relatively lower presence in the dataset. Altogether, our findings suggest that assessing existing legislation alone is not enough to answer our question — gauging the effectiveness of policy requires understandings of local context and a more fair dataset.

Gaps

Firstly, we think we can address the issue of class imbalance in our models by refactoring the dataset and sampling cases that we want to appear more frequently in the data. We can then better test the accuracy of our models in predicting some of the minority classes that were previously being overlooked in our analysis. Another gap that we identified was the overall possibility of certain states simply not reporting their hate crimes. Historically, we understand that there were few regulations mandating that states had to report instances, however, now some states have stricter requirements than others. This could bias the dataset with more entries from states that require reporting relative to those who do not, which we think could have been the cause of certain states showing up as top features in our models.

Next Steps

There was a large amount of legislation that was passed after 2019, however, our dataset did not have data for after this point. If we could expand our dataset to include hate crimes after 2019, we could gauge how this significant increase in hate crime legislation impacted the number of incidents and previously observed trends with features (e.g., bias categories). Additionally, the time horizon for when the levels of legislation we assigned per state varies. With our current dataset, we could also be able to study state trends before and after the enactment of existing legislation (e.g., the number of annual incidents, motivation).

Some next steps would be to try and find more granular information on legislation passed across states so that we can further understand legislation impact on hate crime occurrence. Also, a goal could be to improve the accuracy of our current models by refactoring the dataset and sampling some of the cases we want to emphasize more in the dataset. This would allow us to see if shifting the dataset balance helps train the model better and get better understanding of what features may drive that specific hate crime case. Since we are working to understand how impactful legislation is in reducing hate crimes, we may discover what type of legislation is important for states to include, however, this may shift the focus away from all other types of hate crime biases.